

- 3 Baculovirus is a virus that kills some of the insect pests of major crops. When the virus is released to the outside of the insect body, it is contained within stable, protective structures known as polyhedrons.

The main component of the polyhedron is a protein molecule, polyhedrin.

Polyhedrons can be sprayed onto plants as a bioinsecticide. They are ingested by feeding insect larvae and once inside the insect gut they break down to release the virus.

- (a) Explain why the term infectious disease can be used to describe the effect of baculovirus on insects.

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- (b) Suggest **and** explain the conditions present in the insect gut that can cause the breakdown of polyhedrons.

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- (c) A polyhedrin molecule is composed of three identical polypeptides. Each polypeptide is 245 amino acids long. The first 10 amino acids of the polypeptide are shown in Fig. 3.1.

1 2 3 4 5 6 7 8 9 10
met — pro — asp — tyr — ser — tyr — arg — pro — thr — ile —

Fig. 3.1

Fig. 3.2 **A** is a ribbon diagram of a single polypeptide. Fig. 3.2 **B** shows a polyhedrin molecule composed of 3 polypeptides.

The two diagrams are not to the same scale.

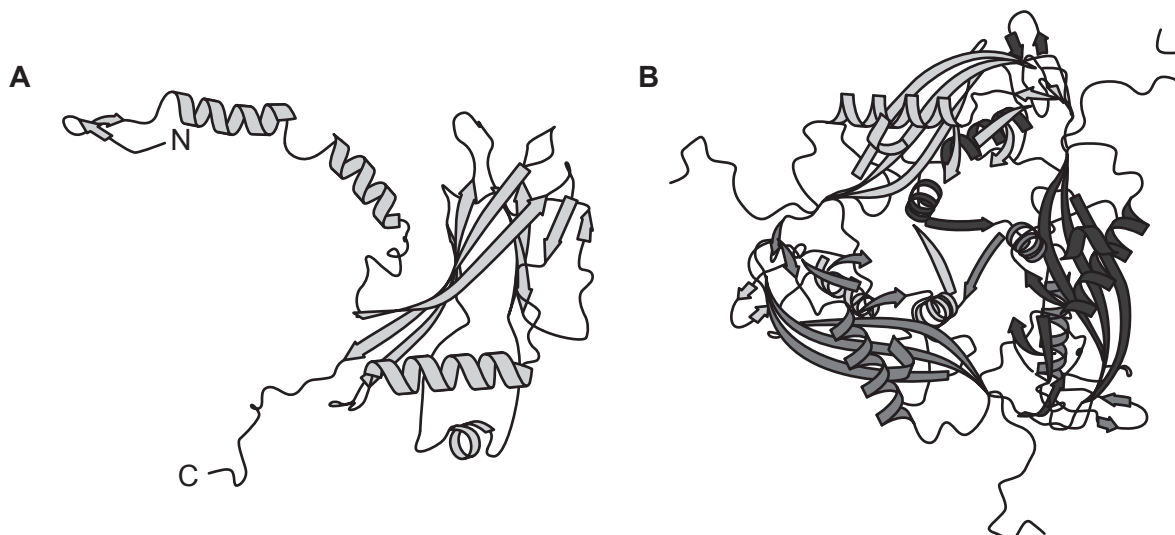


Fig. 3.2

With reference to Fig. 3.1 and Fig. 3.2, describe the structure of a polyhedrin molecule.

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- (d) The sequence of DNA nucleotides for the gene in baculovirus that codes for the polyhedrin polypeptide has been determined.

Explain why the amino acid sequence of the polypeptide **cannot** be used to deduce this sequence of nucleotides in the gene that codes for the polyhedrin polypeptide.

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[Total: 11]